12.2.3 Pie Diagram

The Pie diagram is a circular diagram. A circle which has 360° is divided into different sectors. Angles of the sectors, subtending at the center, are proportional to the magnitudes of the frequency of the components. When a circle is sub-divided into various component sectors, it is called a Pie diagram.

How to construct a pie diagram?

The following procedure can be followed to draw a Pie diagram for a given data:

1. Draw a circle by using a particular radius.
2. The proportional values of various components should be converted into degrees.
3. The angles are obtained by the following formula

$$Angle value = \frac{Value\, of\, component}{Total\, value} \times 360$$

Example 12.5

Construct a Pie diagram to represent the data given in **Table 12.5.**

Table 12.5 Percentage of area under wheat cultivation

Continent	Area in %
North America	12.7
South America	4.07
Europe	25.75
Asia	46.85
Oceania	6.25
Africa	4.38

Continent	Area in %	Calculation	Degree
North America	12.7	$\frac{12.7}{100} \times 360^{\circ} = 45.72$	46°
South America	4.07	$\frac{4.07}{100} \times 360^{\circ} = 14.65$	15°
Europe	25.75	$\frac{25.75}{100} \times 360^{\circ} = 92.7$	92°
Asia	46.85	$\frac{46.85}{100} \times 360^{\circ} = 168.66$	169°
Oceania	6.25	$\frac{6.25}{100} \times 360^{\circ} = 22.5$	22°
Africa	4.38	$\frac{4.38}{100} \times 360^{\circ} = 15.76$	16°
Total	100		360°

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